# 314707003_4.25.R

## CDH

## 2026-04-05

```
rm(list=ls())
library(POE5Rdata)
```

```
##
## 載入套件：'POE5Rdata'
```

```
## 下列物件被遮斷自 'package:datasets':
##
##     euro
```

```
#install.packages("tseries")
library(tseries)
```

```
## Warning: 套件 'tseries' 是用 R 版本 4.5.3 來建造的
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
#problem4.25
mod1 = lm(log(price) ~ sqft, data = collegetown)
mod2 = lm(log(price) ~ log(sqft), data = collegetown)
mod3 = lm(price ~ sqft, data = collegetown)

#(a)
summary(mod1)
```

```
##
## Call:
## lm(formula = log(price) ~ sqft, data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.32528 -0.19199  0.00122  0.21678  0.86609
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.393866   0.043288  101.50   <2e-16 ***
## sqft        0.036044   0.001486   24.26   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.34 on 498 degrees of freedom
## Multiple R-squared:  0.5417, Adjusted R-squared:  0.5408
## F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
# slope and elasticity at sample mean
slope_a = coef(summary(mod1))[[2]] * mean(collegetown$price)
elas_a = coef(summary(mod1))[[2]] * mean(collegetown$sqft)
cat(
  "(a)" ,"\n",
  "slope at the sample mean is : ",slope_a, "\n",
  "elasticity at the sample mean is : ",elas_a,"\n",
  sep=""
)
```

```
## (a)
## slope at the sample mean is : 9.019661
## elasticity at the sample mean is : 0.9833701
```

```
#(b)
summary(mod2)
```

```
##
## Call:
## lm(formula = log(price) ~ log(sqft), data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.36864 -0.20849  0.00487  0.23204  1.21099
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.04965    0.15797   12.97   <2e-16 ***
## log(sqft)    1.02483    0.04839   21.18   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3643 on 498 degrees of freedom
## Multiple R-squared:  0.4738, Adjusted R-squared:  0.4728
## F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16
```

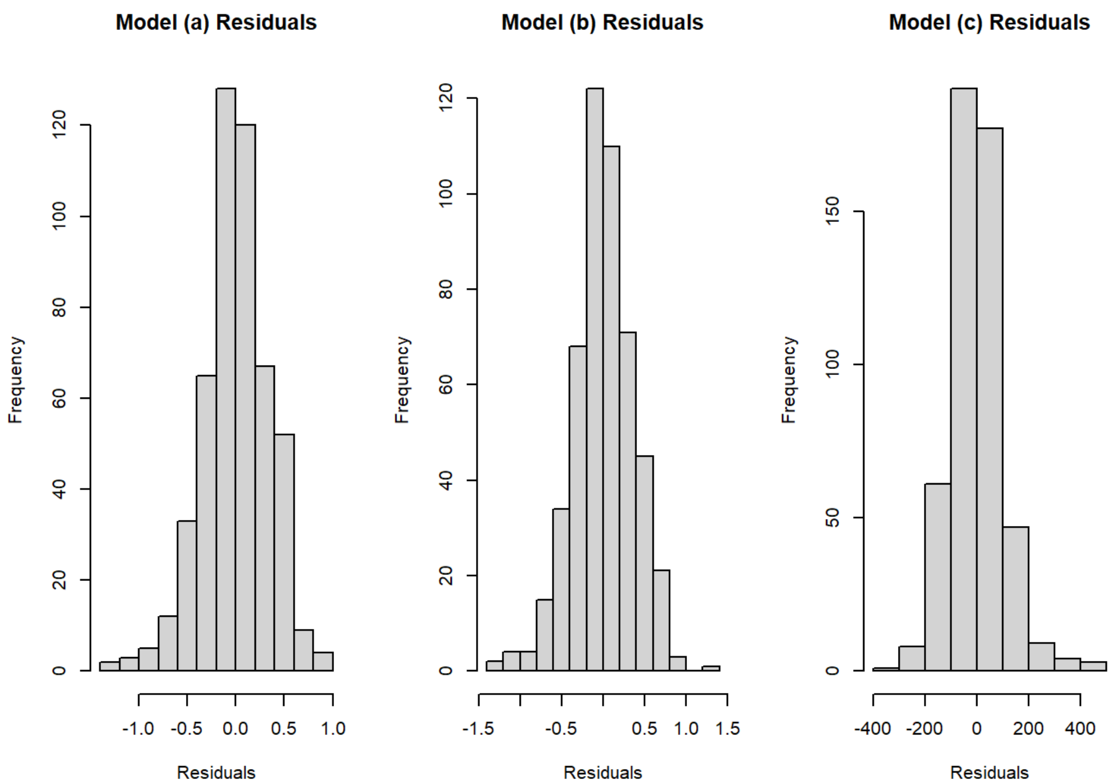
```
# slope and elasticity at sample mean
slope_a = coef(summary(mod2))[[2]] * (mean(collegetown$price) / mean(colleget
own$sqft))
elas_a = coef(summary(mod2))[[2]]
cat(
  "(b)" ,"\n",
  "slope at the sample mean is : ",slope_a, "\n",
  "elasticity at the sample mean is : ",elas_a,"\n",
  sep=""
)
```

```
## (b)
## slope at the sample mean is : 9.399923
## elasticity at the sample mean is : 1.024828
```

```
#(c)
cat(
  "(c)" ,"\n",
  "R-squared value from the linear model is : ",summary(mod3)$r.squared,
"\n",
  "Generalized R-squared from (b) is : ",cor(collegetown$price, exp(fitted(mo
d2)))^2,"\n",
  "Generalized R-squared from (c)(the linear model) is : ",cor(collegetown$pr
ice, fitted(mod3))^2,"\n",
  sep=""
)
```

```
## (c)
## R-squared value from the linear model is : 0.6413167
## Generalized R-squared from (b) is : 0.6445084
## Generalized R-squared from (c)(the linear model) is : 0.6413167
```

```
#(d)
par(mfrow = c(1, 3))
hist(resid(mod1), main = "Model (a) Residuals", xlab = "Residuals")
hist(resid(mod2), main = "Model (b) Residuals", xlab = "Residuals")
hist(resid(mod3), main = "Model (c) Residuals", xlab = "Residuals")
```




```
cat(
  "(d)" ,"\n",
  "the Jarque-Bera statistics for (a) is : ",jarque.bera.test(resid(mod1))
[[1]], "\n",
  "the Jarque-Bera statistics for (b) is : ",jarque.bera.test(resid(mod2))
[[1]],"\n",
  "the Jarque-Bera statistics for (c) is : ",jarque.bera.test(resid(mod3))
[[1]],"\n",
  sep=""
)
```

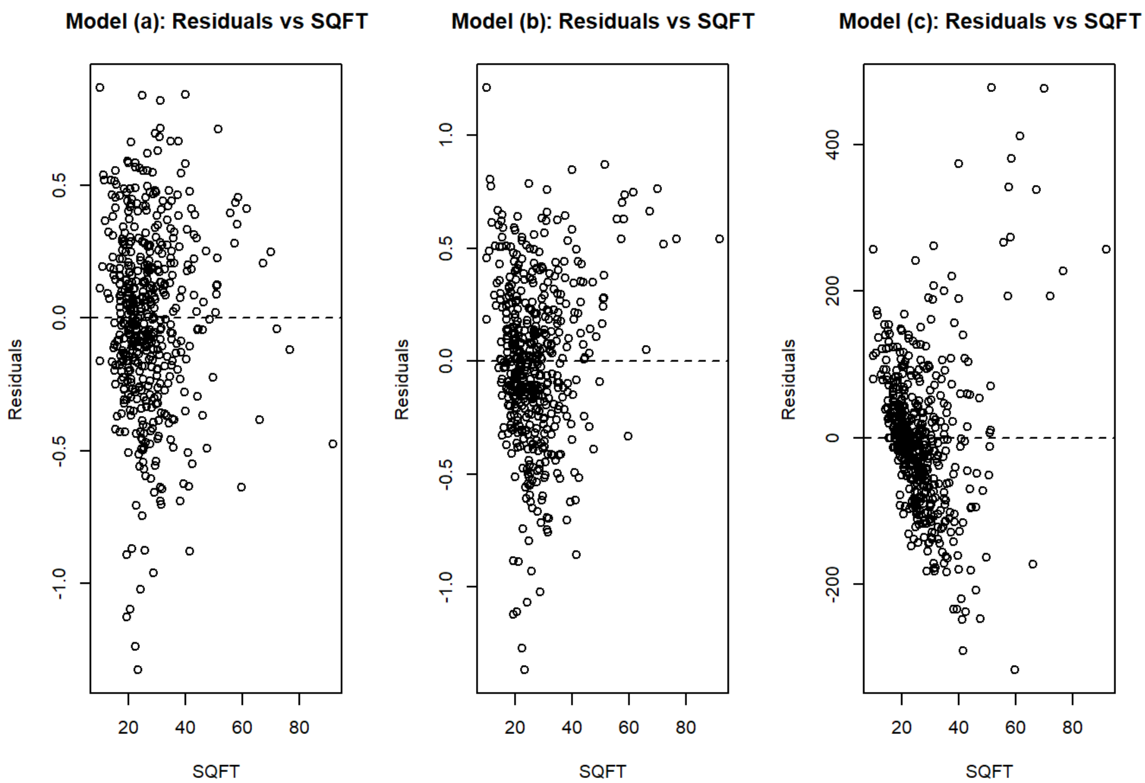
```
## (d)
## the Jarque-Bera statistics for (a) is : 26.67899
## the Jarque-Bera statistics for (b) is : 14.27871
## the Jarque-Bera statistics for (c) is : 221.1115
```

```
cat(
  "(d)" ,"\n",
  "p-value for (a) is : ",jarque.bera.test(resid(mod1))[[3]], "\n",
  "p-value for (b) is : ",jarque.bera.test(resid(mod2))[[3]],"\n",
  "p-value for (c) is : ",jarque.bera.test(resid(mod3))[[3]],"\n",
  sep=""
)
```

```
## (d)
## p-value for (a) is : 1.609651e-06
```

```
## p-value for (b) is : 0.0007932632
## p-value for (c) is : 0
```

```
#(e)
par(mfrow = c(1, 3))
plot(collegetown$sqft, resid(mod1),
     main = "Model (a): Residuals vs SQFT",
     xlab = "SQFT", ylab = "Residuals")
abline(h = 0, lty = 2)
plot(collegetown$sqft, resid(mod2),
     main = "Model (b): Residuals vs SQFT",
     xlab = "SQFT", ylab = "Residuals")
abline(h = 0, lty = 2)
plot(collegetown$sqft, resid(mod3),
     main = "Model (c): Residuals vs SQFT",
     xlab = "SQFT", ylab = "Residuals")
abline(h = 0, lty = 2)
```




```
#(f)
newhouse <- data.frame(sqft = 27) #單位是千美元
cat(
  "(f)" ,"\n",
  "predict value for model (a) is : ",exp(predict(mod1, newhouse))*1000,
"\n",
  "predict value for model (b) is : ",exp(predict(mod2, newhouse))*1000,"\n",
  "predict value for model (c) is : ",predict(mod3, newhouse)*1000,"\n",
```

```
  sep=""
)
```

```
## (f)
## predict value for model (a) is : 214233.6
## predict value for model (b) is : 227538.6
## predict value for model (c) is : 246455.7
```

```
#(g)
pi_a = exp(predict(mod1, newdata = newhouse, interval = "prediction", level =
0.95))
pi_b = exp(predict(mod2, newdata = newhouse, interval = "prediction", level =
0.95))
pi_c = predict(mod3, newdata = newhouse, interval = "prediction", level = 0.9
5)
cat(
  "(f): 95%PI" ,"\n",
  "model (a) log-linear: ",pi_a[[2]],",",pi_a[[3]], "\n",
  "model (b) log-log: ",pi_b[[2]],",",pi_b[[3]],"\n",
  "model (c) linear : ",pi_c[[2]],",",pi_c[[3]],"\n",
  sep=""
)
```

```
## (f): 95%PI
## model (a) log-linear: 109.7685,418.1167
## model (b) log-log: 111.1406,465.8406
## model (c) linear : 44.27727,448.6341
```